Cybersecurity Hands-On Lab Report

Module 2: Network Discovery and Scanning with Nmap
Instructor: Lebian Wilfried NIKIEMA, Ph.D
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Student: [SEMPORE JERIEL WENDINDA]

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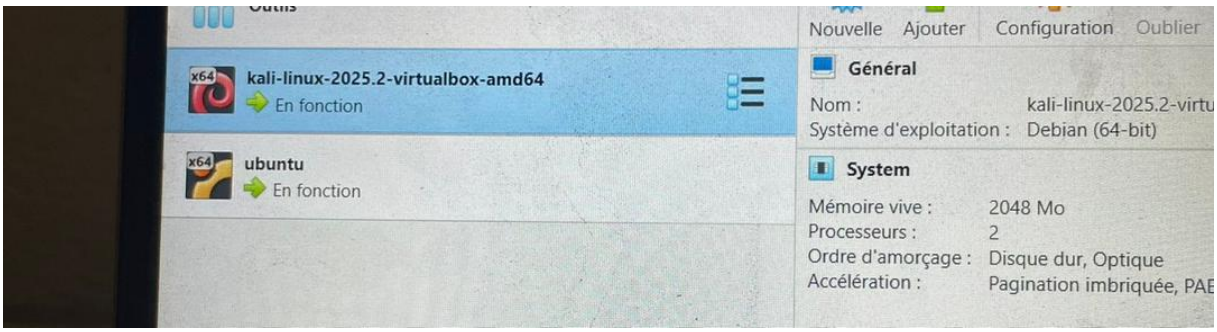
1. Introduction

This lab focused on practicing network reconnaissance, which is the first phase of penetration testing. Using Kali Linux as the attacker machine and Ubuntu as the target, we performed several scans with Nmap to discover active hosts, identify open ports, detect services and versions, and analyze the operating system.



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2. Commands and Results

Part 1: Verify Installation

Command:

nmap --version

Result:

Nmap version 7.98 (https://nmap.org)

Platform: x86_64-pc-linux-gnu

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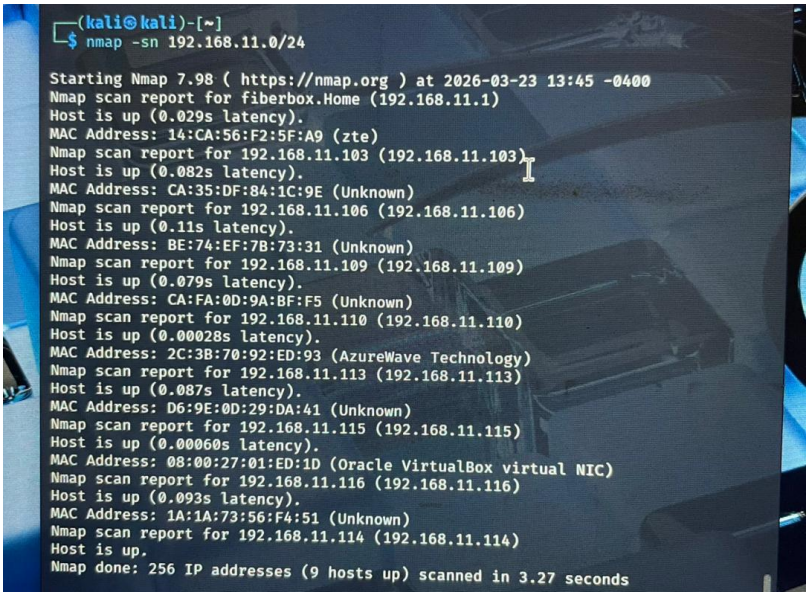
Part 2: Discover Active Hosts

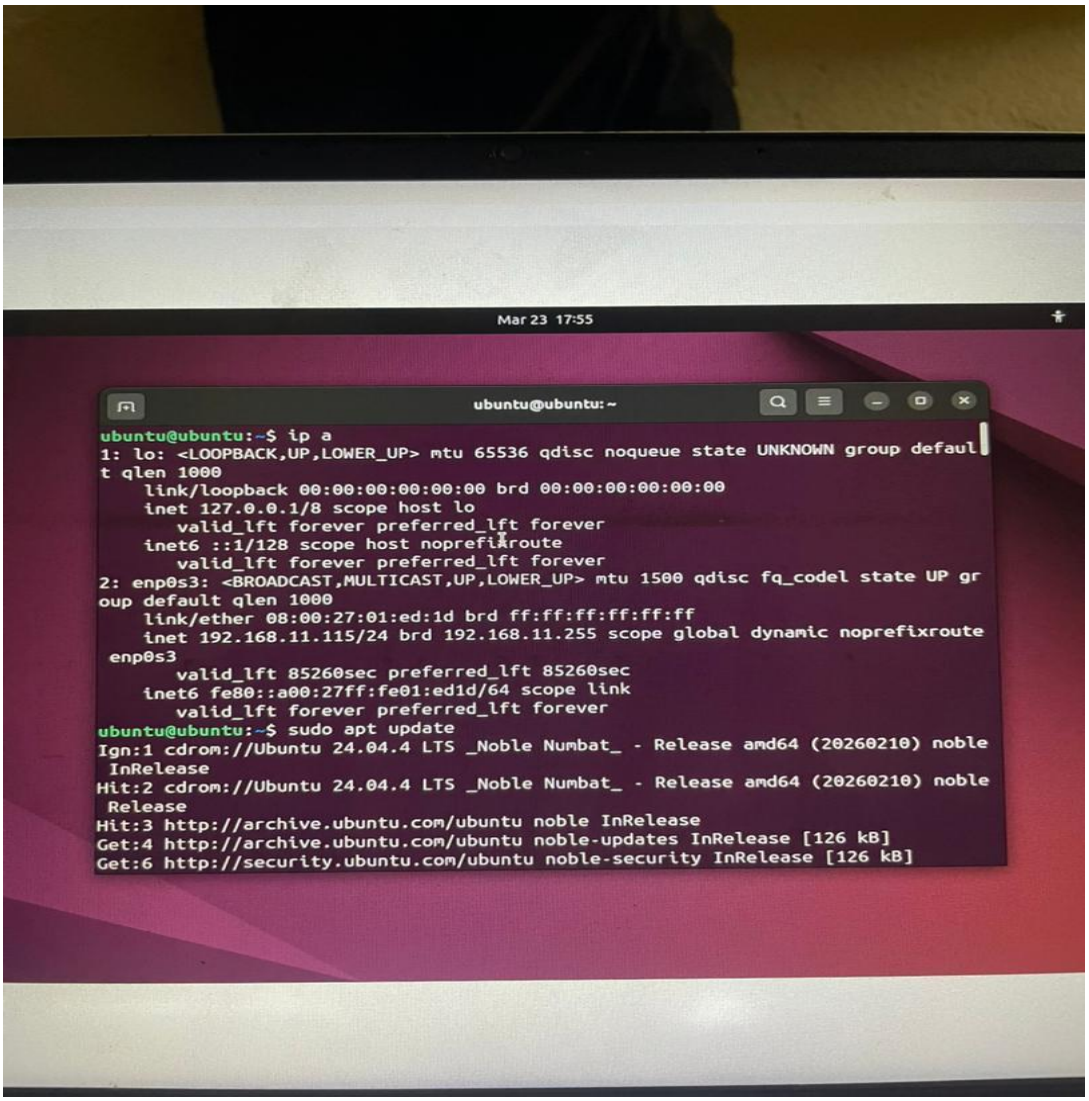
Command:

nmap -sn 192.168.11.0/24

Result:
Nine hosts detected, including:

- 192.168.11.1 (Router – Fiberbox)
- 192.168.11.115 (Ubuntu – Oracle VirtualBox NIC)





Part 3: Scan Open Ports

Command:

nmap 192.168.11.115

Result:

PORT STATE SERVICE

22/tcp open ssh

80/tcp open http

443/tcp open https

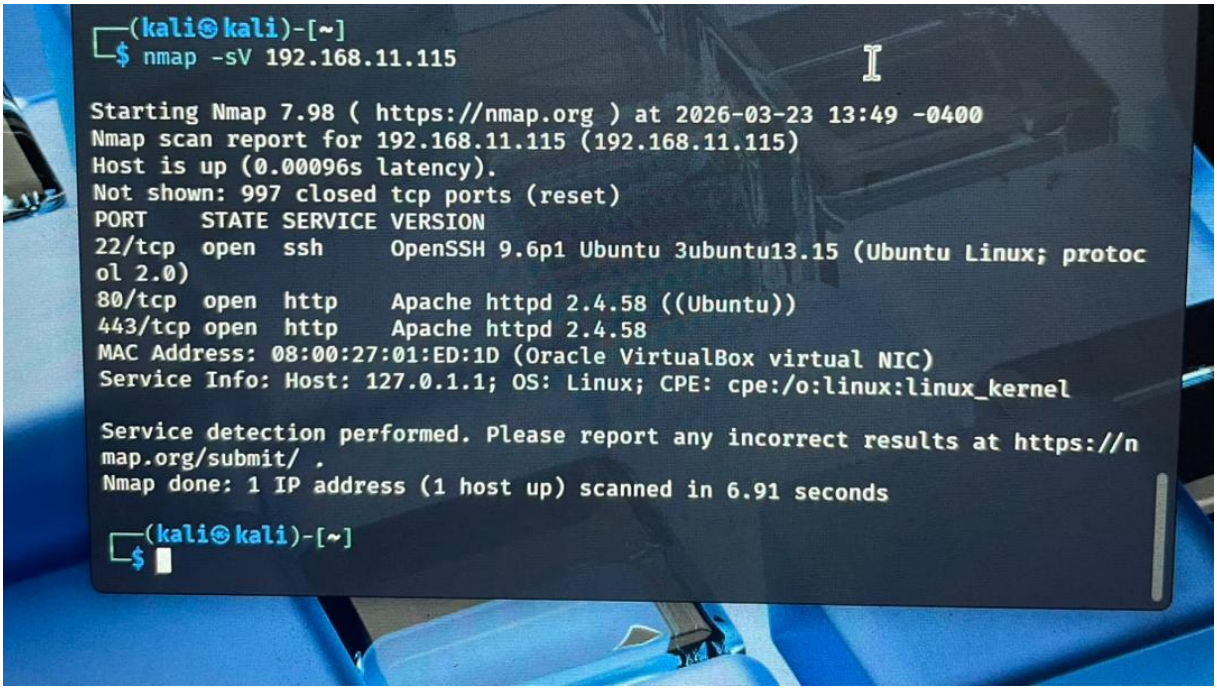
Part 4: Service Detection

Command:

nmap -sV 192.168.11.115

Result:

- SSH → OpenSSH 9.6p1 Ubuntu
- HTTP → Apache httpd 2.4.58 (Ubuntu)
- HTTPS → Apache httpd 2.4.58 with SSL



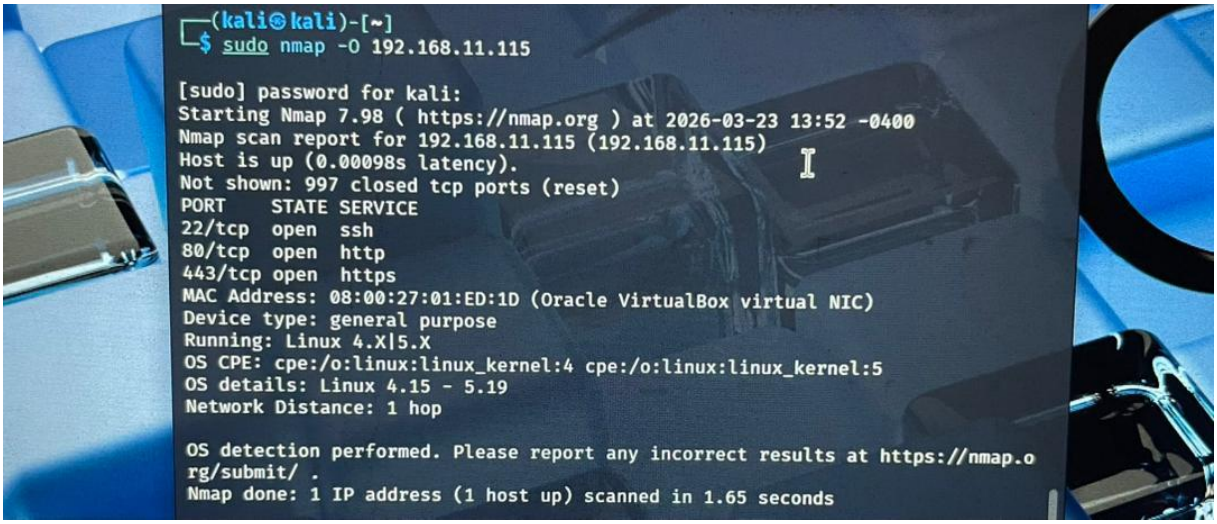
Part 5: OS Detection

Command:

sudo nmap -O 192.168.11.115

Result:

- Detected OS: Linux kernel 4.15 – 5.19



Part 6: Aggressive Scan

Command:

sudo nmap -A 192.168.11.115

Result:

- SSH keys identified
- Apache default page detected
- OS confirmed: Linux kernel 4.15 – 5.19

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┌──(kali㉿kali)-[~]
└─$ sudo nmap -A 192.168.11.115

Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-23 13:53 -0400
Nmap scan report for 192.168.11.115 (192.168.11.115)
Host is up (0.00094s latency).
Not shown: 997 closed tcp ports (reset)
PORT    STATE SERVICE VERSION
22/tcp  open  ssh     OpenSSH 9.6p1 Ubuntu 3ubuntu13.15 (Ubuntu Linux; protoc
ol 2.0)
| ssh-hostkey:
|   256 0e:d9:63:32:a6:8e:4b:5d:6a:b1:62:1d:b6:d8:0e:0a (ECDSA)
|_  256 98:9b:4f:34:5a:40:19:a4:38:31:ee:9c:ed:26:30:10 (ED25519)
80/tcp  open  http    Apache httpd 2.4.58 ((Ubuntu))
|_http-server-header: Apache/2.4.58 (Ubuntu)
|_http-title: Apache2 Ubuntu Default Page: It works
443/tcp open  http    Apache httpd 2.4.58
|_http-title: Apache2 Ubuntu Default Page: It works
|_http-server-header: Apache/2.4.58 (Ubuntu)
MAC Address: 08:00:27:01:ED:1D (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.19
Network Distance: 1 hop
Service Info: Host: 127.0.1.1; OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE
HOP RTT     ADDRESS
1   0.94 ms 192.168.11.115 (192.168.11.115)

OS and Service detection performed. Please report any incorrect results at ht
tps://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.06 seconds
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3. Analysis Questions

1. How many hosts were detected? → 9 hosts (including Ubuntu at 192.168.11.115).
2. Which ports are most common? → 22 (SSH), 80 (HTTP), 443 (HTTPS).
3. Why are open ports a risk? → They allow attackers to attempt connections or exploit vulnerabilities.
4. What information can attackers gather? → Service versions, operating system details, and network configuration.

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4. Security Observations

- SSH open → risk of brute force attacks if weak passwords are used.
- HTTP open → risk of data exposure if not secured.
- HTTPS open → safer, but depends on SSL configuration.
- More exposed services increase the attack surface.

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5. Conclusion

This lab demonstrated the use of Nmap for network reconnaissance. We successfully discovered active hosts, identified open ports, detected services and versions, and analyzed the operating system. The exercise highlights that reconnaissance is the first step in cyberattacks and shows the importance of limiting open ports to reduce risks.

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6. Suggestions

- Enable a firewall on Ubuntu to restrict access to unnecessary ports.
- Use strong passwords for SSH.
- Configure HTTPS with valid certificates to secure communication.
- Regularly update services (Apache, OpenSSH) to patch known vulnerabilities.